

Adolescent Physical Activity & Depressive Affect Inference

ONE-WAY ANOVA & TUKEY HSD POST-HOC TEST

FINAL REPORT

STUDY CONTEXT

RESEARCH QUESTION

Does weekly physical activity frequency significantly affect feelings of sadness / hopelessness in US high school students?

SAMPLE PROFILE (YRBS 2007)

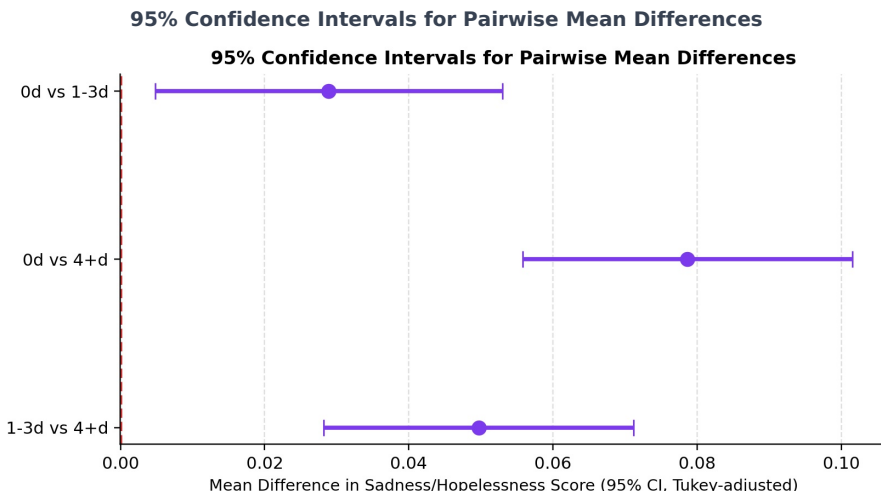
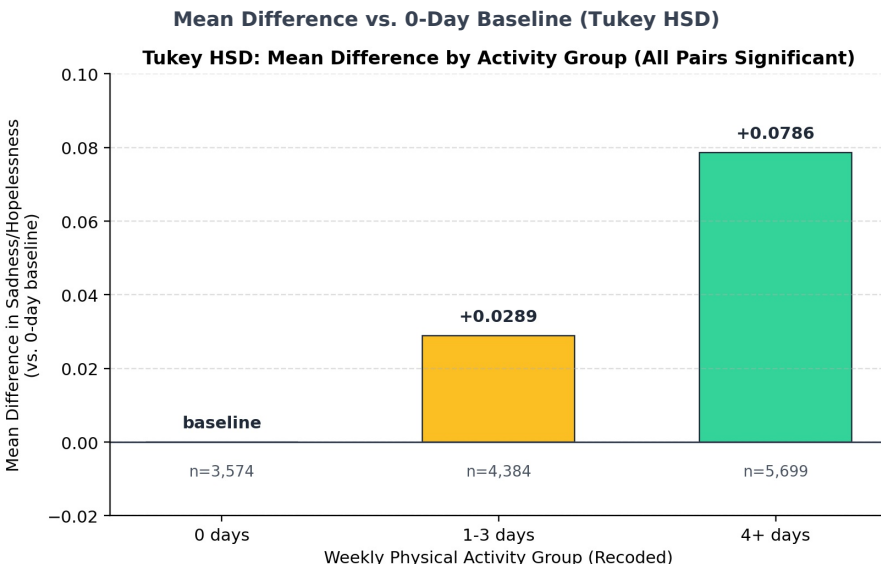
GROUP	N-SIZE
0 days	3,574
1-3 days	4,384
4+ days	5,699
Total Valid N	13,657

LARGEST MEAN DIFFERENCE

0 days vs. 4+ days: +0.0786

Note: SadOrHopeless is a binary YRBS indicator (felt sad/hopeless for ≥ 2 weeks in past 12 months). PhysicalActivity5OrMoreDays was recoded into three exercise-frequency groups. Analysis excludes missing values (N = 13,657).

VISUAL ANALYSIS



INFERENCE RESULTS

P-VALUE (ANOVA)

5.74×10^{-16}

Threshold $\alpha = 0.05$

F-STATISTIC

35.18

df = (2, 13654)

Test: One-Way ANOVA (OLS)

Sum of Sq. (between): 14.69

Residual Sum of Sq.: 2850.83

GROUP 1	GROUP 2	MEAN DIFF	P-ADJ	REJECT
0 days	1-3 days	0.0289	0.0139	True
0 days	4+ days	0.0786	0.0000	True
1-3 days	4+ days	0.0497	0.0000	True

VERDICT: REJECT H_0

A statistically significant difference exists across activity groups ($p < 0.001$). Tukey HSD confirms all three pairwise comparisons are significant: students who exercise more frequently report systematically lower sadness/hopelessness scores, with the largest gap between 0-day and 4+ day groups.